

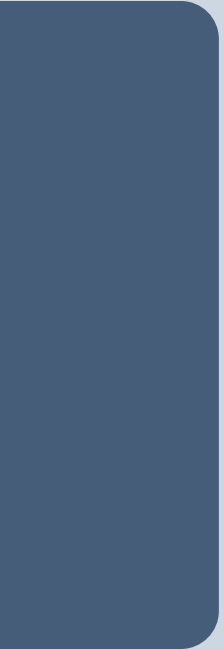


Reto Técnico de Evaluación

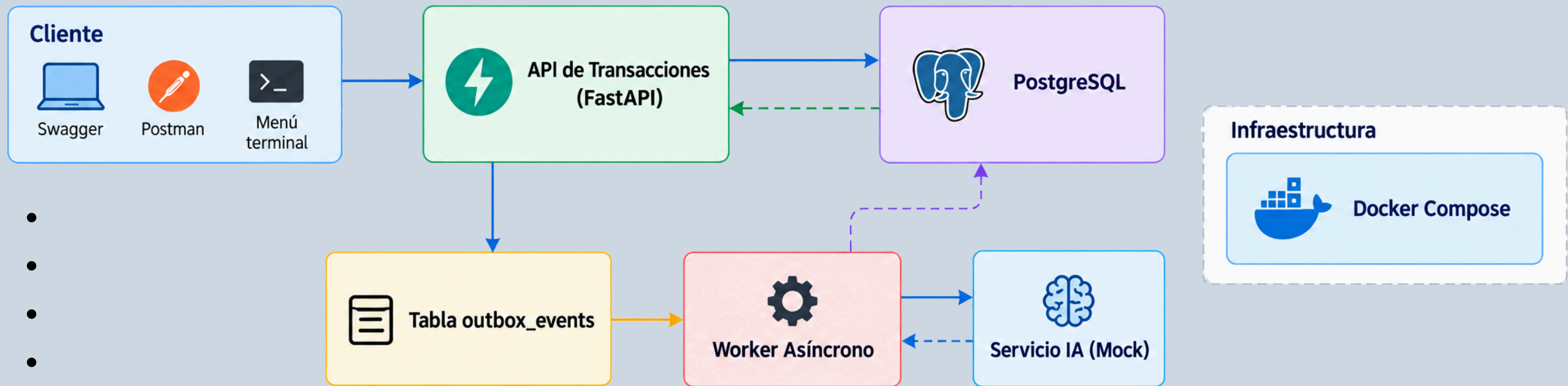
"SmartBancs App"

**Arquitectura, Desarrollo Práctico,
Operación, IA y Observabilidad**

Alexander Vera



Solución – Arquitectura



Infraestructura, Base de Datos y Desarrollo del Backend

SmartBancs Transaction API 0.1.0 OAS 3.1

/openapi.json

default

GET

/health Health

GET

/metrics Metrics

GET

/accounts/{account_id} Get Account

GET

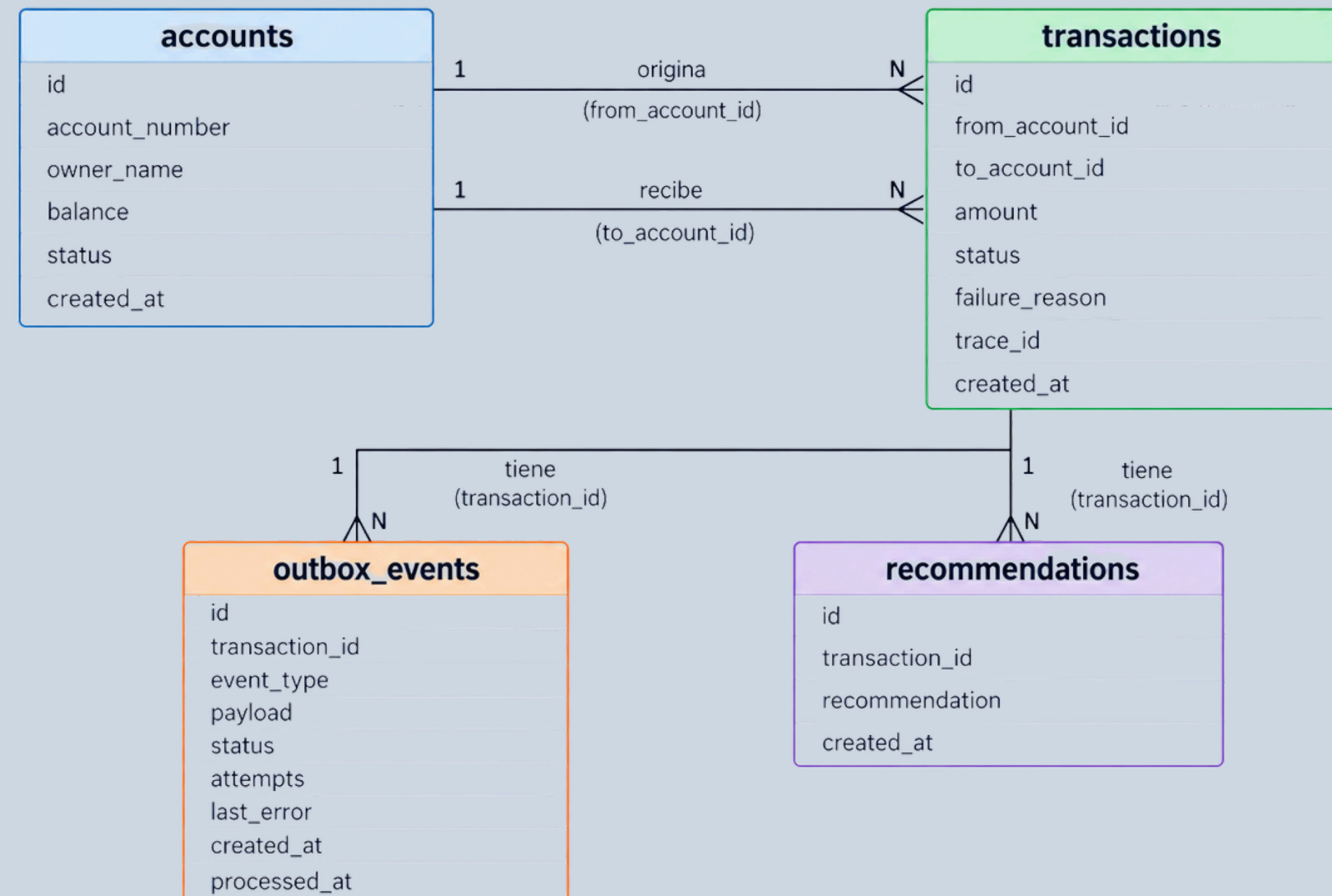
/transactions/{transaction_id} Get Transaction

GET

/transactions/{transaction_id}/processing-status Get Processing Status

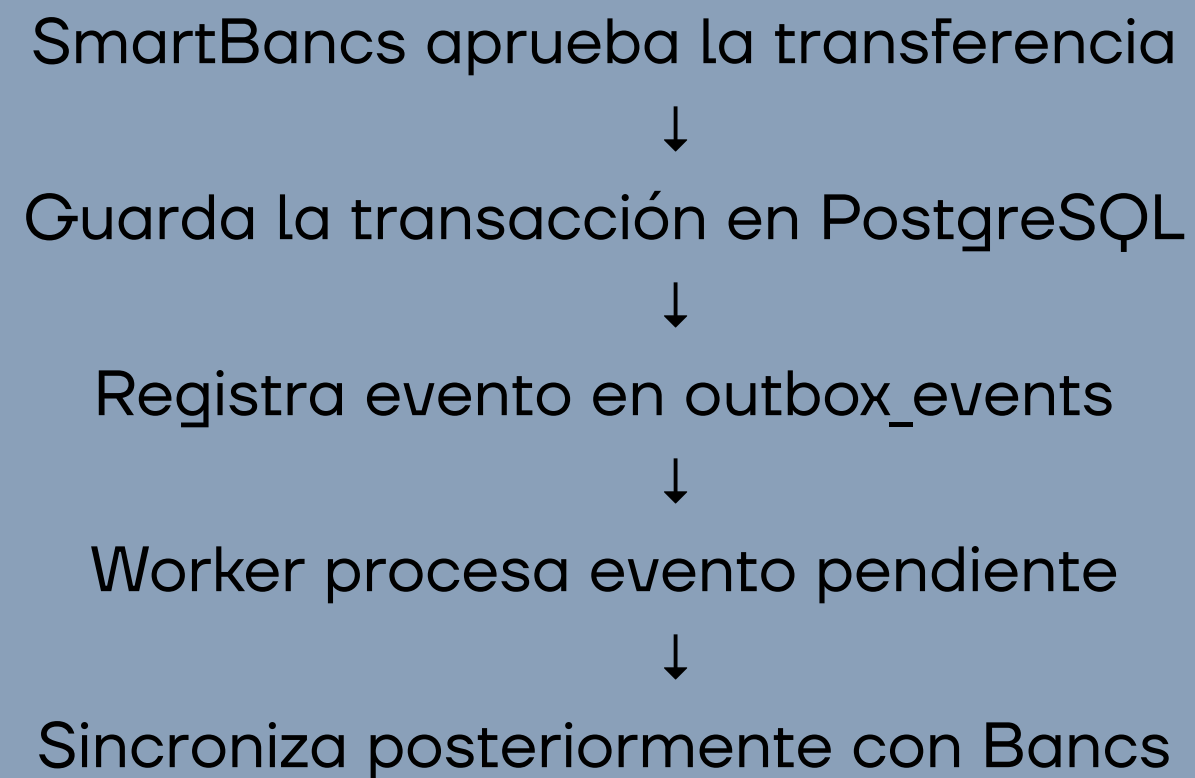
POST

/transactions Create Transaction

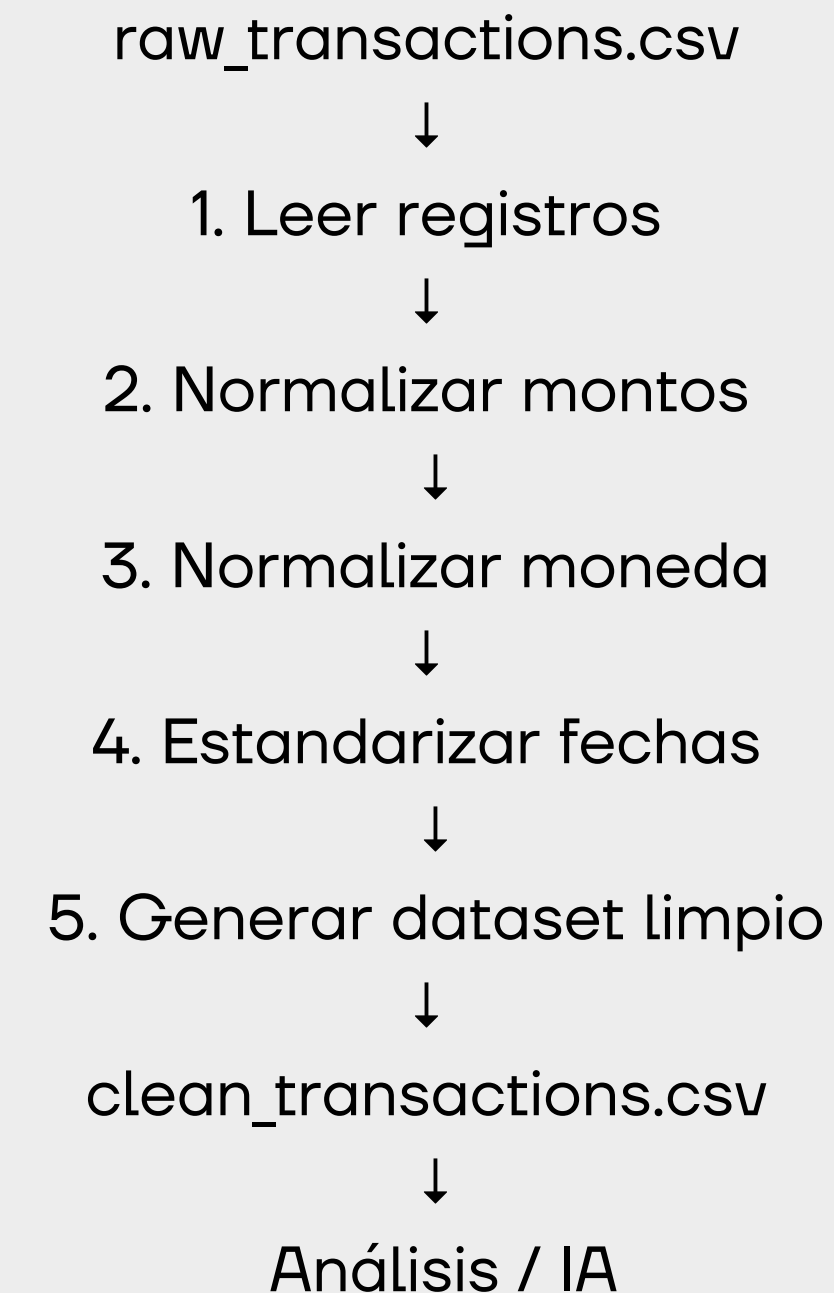


Bancs: Manejo, Utilización e Integración de Datos

Estrategia de sincronización



Transformación de datos



Inteligencia Artificial: Implementación y Despliegue

Modelo de recomendacion financiera

Integración práctica de IA

Transferencia aprobada
↓
Evento en outbox_events
↓
Worker asíncrono
↓
AI Service (Mock)
↓
Genera recomendación
↓
Guarda resultado

**Modelo
propuesto:
Random Forest**

Ciclo de vida del modelo

Datos limpios del ETL
↓
Entrenamiento
↓
Validación
↓
Versionamiento
↓
Despliegue
↓
Monitoreo
↓
Reentrenamiento

Observabilidad

```
ai-service.log X
logs > ai-service.log
1 {"service": "smartbanco-ai-service", "event": "recommendation_created", "trace_id": "42ab61fd-3ebc-47d6-b7df-184dd1d2ca31", "transaction_id": "0e467ae0-1866-4728-b6bb-46715ba8c9a9", "model_version": "v1.0.0"}
2
```

```
api.log X
logs > api.log
1 {"service": "smartbanco-api", "event": "transaction_received", "trace_id": "42ab61fd-3ebc-47d6-b7df-184dd1d2ca31", "from_account_id": 1, "to_account_id": 2, "amount": "100"}
2 {"service": "smartbanco-api", "event": "accounts_locked", "trace_id": "42ab61fd-3ebc-47d6-b7df-184dd1d2ca31", "account_ids": [1, 2]}
3 {"service": "smartbanco-api", "event": "balances_updated", "trace_id": "42ab61fd-3ebc-47d6-b7df-184dd1d2ca31", "from_account_id": 1, "to_account_id": 2, "amount": "100"}
4 {"service": "smartbanco-api", "event": "outbox_event_created", "trace_id": "42ab61fd-3ebc-47d6-b7df-184dd1d2ca31", "transaction_id": "0e467ae0-1866-4728-b6bb-46715ba8c9a9", "event_type": "TRANSACTION"}
5 {"service": "smartbanco-api", "event": "transaction_approved", "trace_id": "42ab61fd-3ebc-47d6-b7df-184dd1d2ca31", "transaction_id": "0e467ae0-1866-4728-b6bb-46715ba8c9a9", "duration_ms": 21}
6 {"service": "smartbanco-api", "event": "transaction_received", "trace_id": "0f0d0163-cbcd-4a87-913b-f5f90adff652", "from_account_id": 3, "to_account_id": 2, "amount": "500"}
7 {"service": "smartbanco-api", "event": "accounts_locked", "trace_id": "0f0d0163-cbcd-4a87-913b-f5f90adff652", "account_ids": [2, 3]}
8 {"service": "smartbanco-api", "event": "transaction_rejected", "trace_id": "0f0d0163-cbcd-4a87-913b-f5f90adff652", "transaction_id": "045319f5-af3d-448f-b634-0a80f87b5676", "reason": "Insufficient funds"}
9
```

```
worker.log X
logs > worker.log
1 {"service": "smartbanco-worker", "event": "worker_started"}
2 {"service": "smartbanco-worker", "event": "outbox_event_processing", "event_id": "9277fa64-ee4b-4065-83f0-7bf9924d3acf", "transaction_id": "0e467ae0-1866-4728-b6bb-46715ba8c9a9", "event_type": "TRANSACTION"}
3 {"service": "smartbanco-worker", "event": "ai_service_called", "trace_id": "42ab61fd-3ebc-47d6-b7df-184dd1d2ca31", "transaction_id": "0e467ae0-1866-4728-b6bb-46715ba8c9a9", "model_version": "v1.0.0"}
4 {"service": "smartbanco-worker", "event": "outbox_event_processed", "event_id": "9277fa64-ee4b-4065-83f0-7bf9924d3acf", "transaction_id": "0e467ae0-1866-4728-b6bb-46715ba8c9a9"}
5
```

Operaciones: Incidente Crítico Simulado

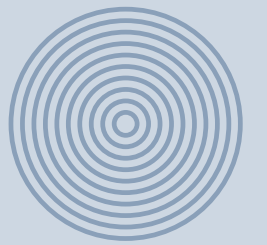
```
api.log X
logs > api.log
1 {"service": "smartbancs-api", "event": "transaction_received", "trace_id": "42ab61fd-3ebc-47d6-b7df-184dd1d2ca31", "from_account_id": 1, "to_account_id": 2, "amount": "100"}
2 {"service": "smartbancs-api", "event": "accounts_locked", "trace_id": "42ab61fd-3ebc-47d6-b7df-184dd1d2ca31", "account_ids": [1, 2]}
3 {"service": "smartbancs-api", "event": "balances_updated", "trace_id": "42ab61fd-3ebc-47d6-b7df-184dd1d2ca31", "from_account_id": 1, "to_account_id": 2, "amount": "100"}
4 {"service": "smartbancs-api", "event": "outbox_event_created", "trace_id": "42ab61fd-3ebc-47d6-b7df-184dd1d2ca31", "transaction_id": "0e467ae0-1866-4728-b6bb-46715ba8c9a9", "event_type": "TRANSACTION"}
5 {"service": "smartbancs-api", "event": "transaction_approved", "trace_id": "42ab61fd-3ebc-47d6-b7df-184dd1d2ca31", "transaction_id": "0e467ae0-1866-4728-b6bb-46715ba8c9a9", "duration_ms": 21}
6 {"service": "smartbancs-api", "event": "transaction_received", "trace_id": "0f0d0163-cbcd-4a87-913b-f5f90adff652", "from_account_id": 3, "to_account_id": 2, "amount": "500"}
7 {"service": "smartbancs-api", "event": "accounts_locked", "trace_id": "0f0d0163-cbcd-4a87-913b-f5f90adff652", "account_ids": [2, 3]}
8 {"service": "smartbancs-api", "event": "transaction_rejected", "trace_id": "0f0d0163-cbcd-4a87-913b-f5f90adff652", "transaction_id": "045319f5-af3d-448f-b634-0a80f87b5676", "reason": "Insufficient"}
9
```

```
metrics.prom X
logs > metrics.prom
1 # HELP python_gc_objects_collected_total Objects collected during gc
2 # TYPE python_gc_objects_collected_total counter
3 python_gc_objects_collected_total{generation="0"} 2071.0
4 python_gc_objects_collected_total{generation="1"} 3842.0
5 python_gc_objects_collected_total{generation="2"} 118.0
6 # HELP python_gc_objects_uncollectable_total Uncollectable objects found during GC
7 # TYPE python_gc_objects_uncollectable_total counter
8 python_gc_objects_uncollectable_total{generation="0"} 0.0
9 python_gc_objects_uncollectable_total{generation="1"} 0.0
10 python_gc_objects_uncollectable_total{generation="2"} 0.0
11 # HELP python_gc_collections_total Number of times this generation was collected
12 # TYPE python_gc_collections_total counter
13 python_gc_collections_total{generation="0"} 138.0
14 python_gc_collections_total{generation="1"} 12.0
15 python_gc_collections_total{generation="2"} 1.0
16 # HELP python_info Python platform information
17 # TYPE python_info gauge
18 python_info{implementation="CPython",major="3",minor="12",patchlevel="14",version="3.12.14"} 1.0
19 # HELP process_virtual_memory_bytes Virtual memory size in bytes.
20 # TYPE process_virtual_memory_bytes gauge
21 process_virtual_memory_bytes 2.79228416e+08
22 # HELP process_resident_memory_bytes Resident memory size in bytes.
23 # TYPE process_resident_memory_bytes gauge
24 process_resident_memory_bytes 7.2740864e+07
25 # HELP process_start_time_seconds Start time of the process since unix epoch in seconds.
26 # TYPE process_start_time_seconds gauge
27 process_start_time_seconds 1.78995892047e+09
28 # HELP process_cpu_seconds_total Total user and system CPU time spent in seconds.
29 # TYPE process_cpu_seconds_total counter
30 process_cpu_seconds_total 1.31
31 # HELP process_open_fds Number of open file descriptors.
32 # TYPE process_open_fds gauge
33 process_open_fds 17.0
34 # HELP process_max_fds Maximum number of open file descriptors.
35 # TYPE process_max_fds gauge
36 process_max_fds 1.048576e+06
37 # HELP transactions_total Total transaction requests
38 # TYPE transactions_total counter
39 transactions_total{status="APPROVED"} 1.0
```

Acciones inmediatas

- Confirmar impacto: revisar metricas, logs y cantidad de errores.
- Identificar si el problema esta en API, PostgreSQL, worker o IA.
- Si hay conexiones bloqueadas, identificar sesiones en PostgreSQL.
- Cancelar consultas bloqueadas o procesos problematicos si es necesario.
- Ajustar temporalmente pool de conexiones o recursos de base de datos.

Operaciones: Gestión de Incidentes TI



Estructura propuesta de post mortem

- resumen del incidente;
- fecha y hora de inicio;
- fecha y hora de cierre;
- impacto en usuarios;
- cantidad de transacciones afectadas;
- servicios afectados;
- causa raíz;
- senales que permitieron detectarlo;
- acciones ejecutadas;
- que funciono;
- que no funciono;
- acciones correctivas;
- responsable;
- fecha compromiso.

Acciones preventivas de infraestructura

- alertas por latencia;
- alertas por errores de base de datos;
- monitoreo de locks/deadlocks;
- revisar pool de conexiones;
- pruebas de carga.

Acciones preventivas de codigo

- mantener bloqueos en orden consistente;
- optimizar consultas;
- agregar índices;
- pruebas de concurrencia;
- monitoreo de eventos PENDING o FAILED.

Muchas gracias

